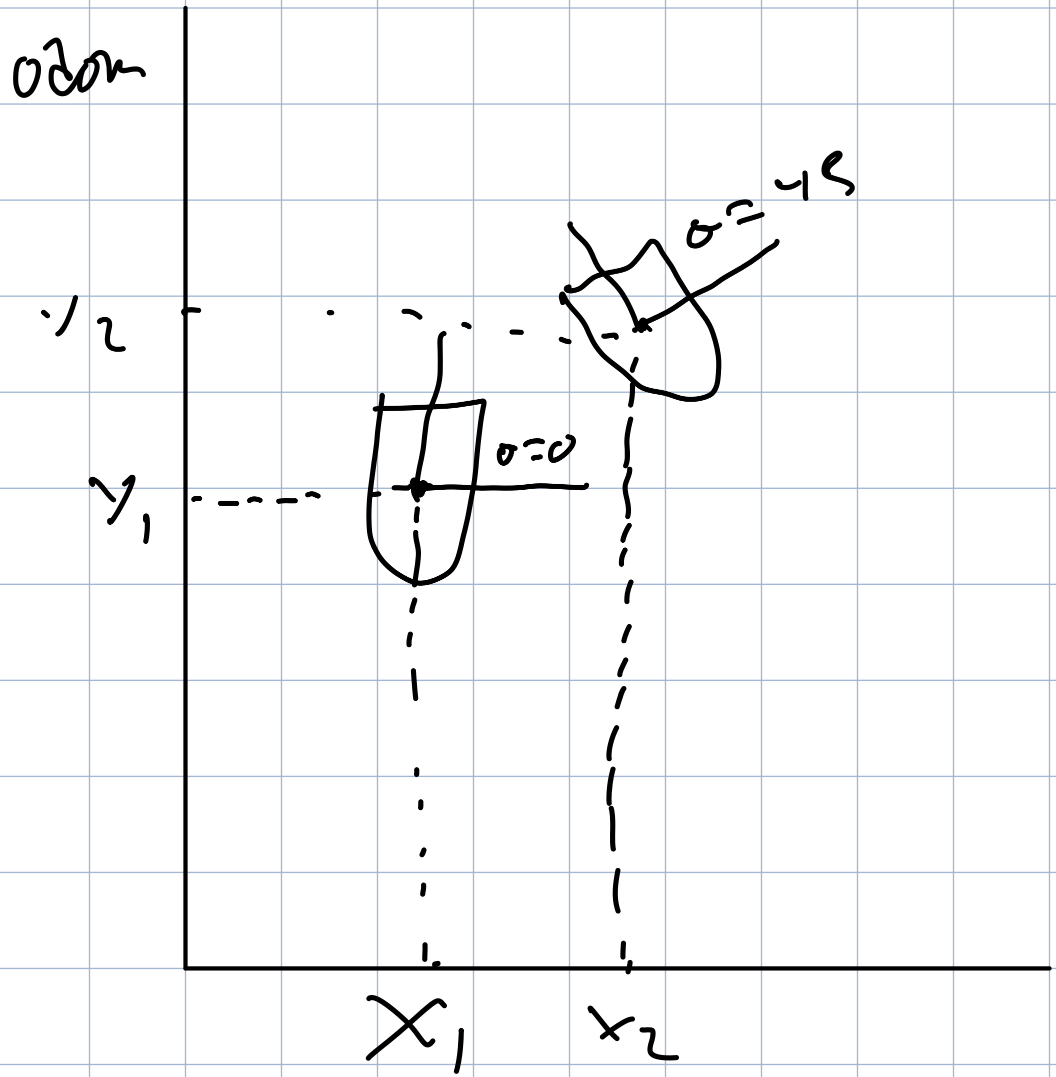
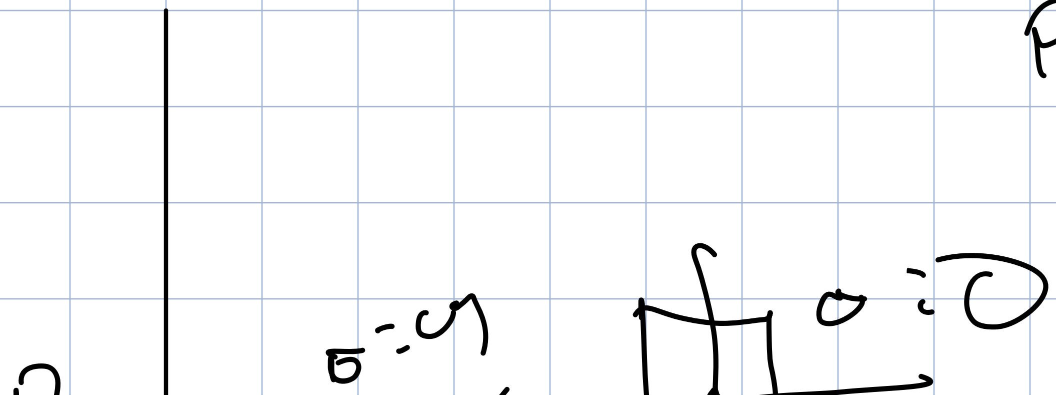
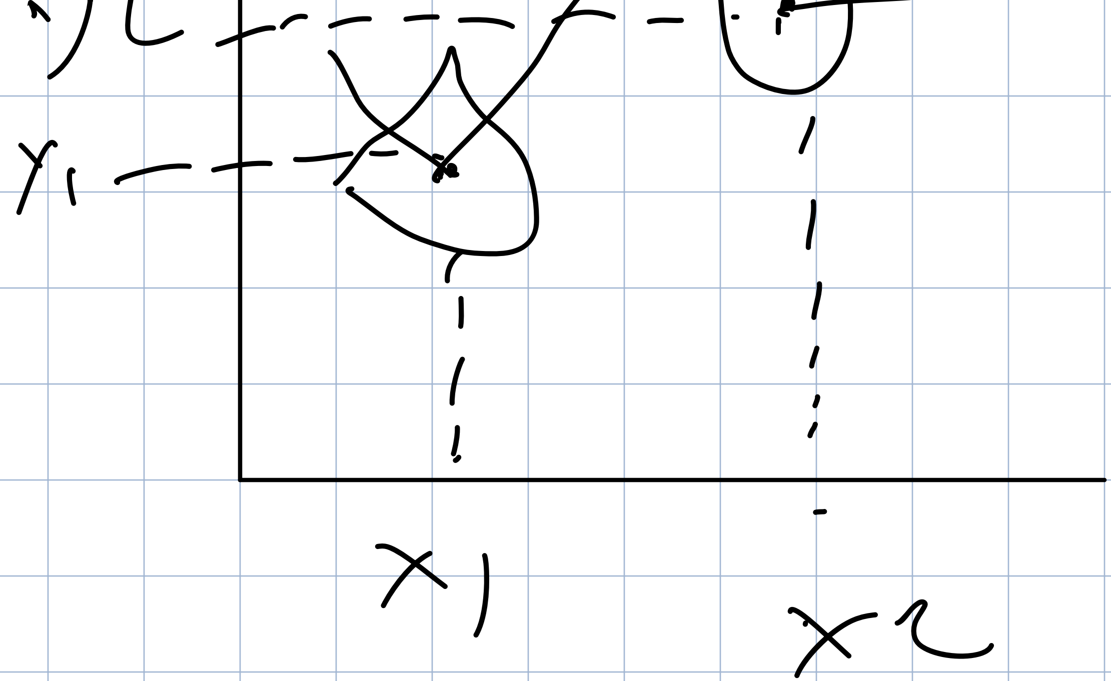

odom
y_2
y_1
$\theta=45$
$\theta=0$
x_1
x_2

$$\left.\begin{aligned}\Delta x &= (x_2-x_1)\\ \Delta y &= (y_2-y_1)\\ \Delta\theta &= (45-0) = 45\end{aligned}\right]$$

Is this not just the change in relative position???





$$\Delta x = (x2 - x1)$$

$$\Delta y = (y2 - y1)$$

$$\Delta \theta = (0 - 45)$$

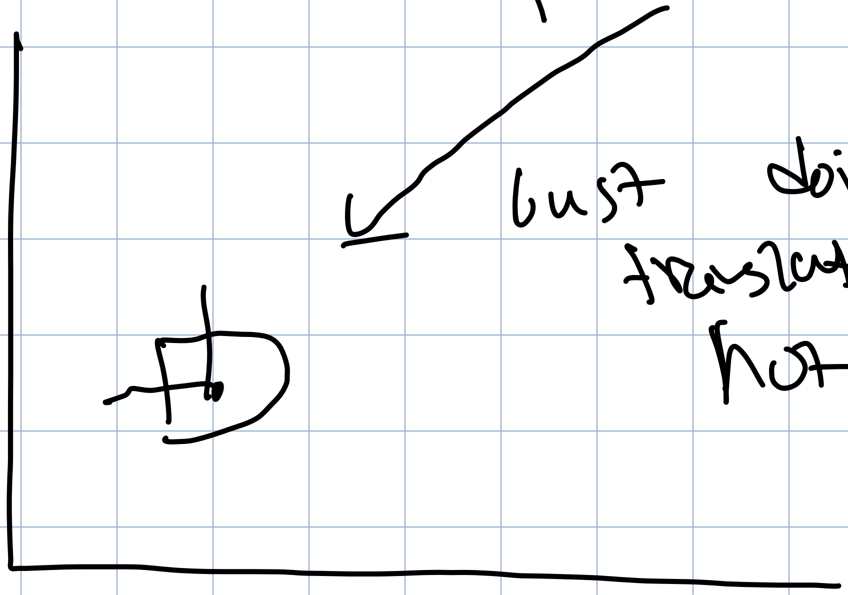
$$= -45$$

Same ???

but I think

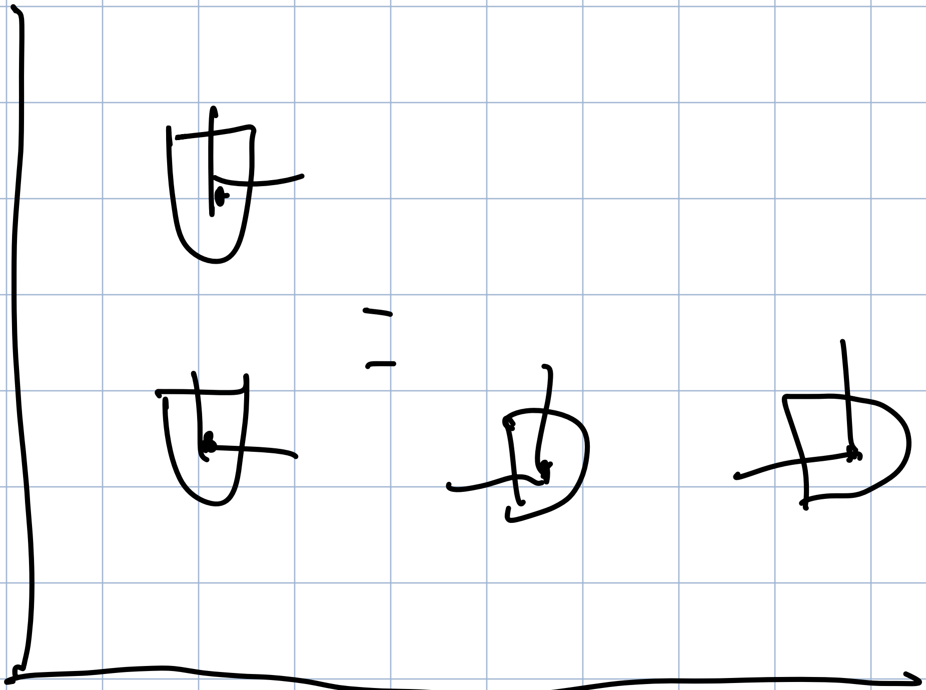
I found the

problem:



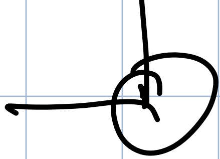
just doing that translation above is not the same for this frame

for example:

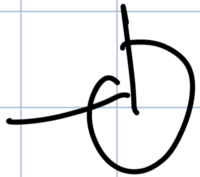


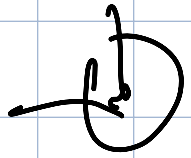
so I guess, maybe assuming that each particle has a steering

$\theta = 0$

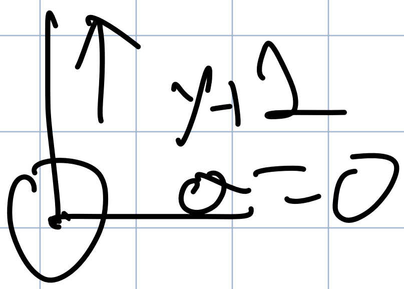


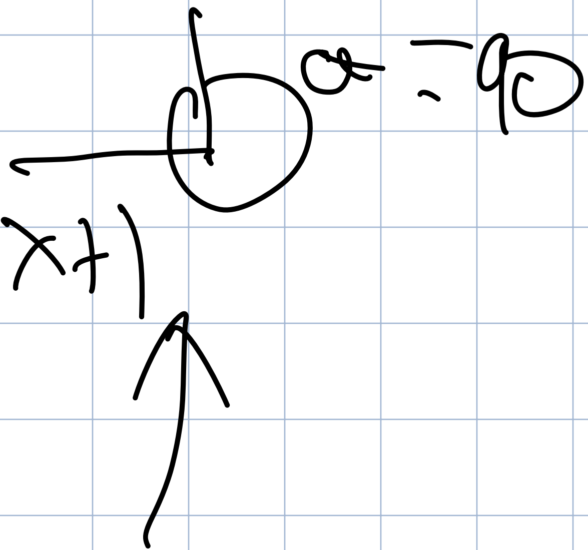
could help





like we can't say particle drive forward, but we need it to move along its heading





$\text{change} \cdot \cos(\theta) = 1$

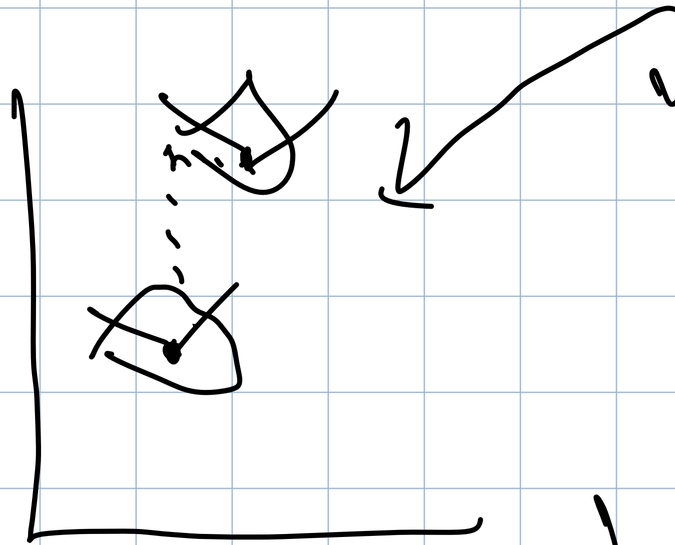
$\text{change} \cdot \sin(\theta) = 0$

$\text{change} \cdot \cos(90) = 0$

$\text{change} \cdot \sin(90) = 1$

So we need to transform odom → self-odom

here's why:

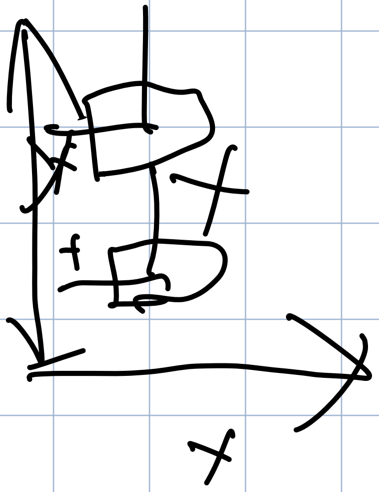


we know the odom change through x, y, theta, but not the personal axis change relative to itself

$o = aO$

$$\begin{bmatrix} o \\ 2 \end{bmatrix}\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$\sin\theta \quad \cos\theta \Big]$

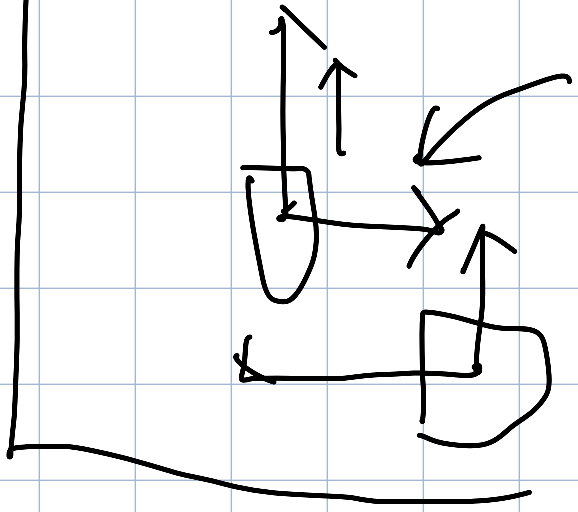


this is wrong? direction

$$\begin{bmatrix} 0\cos\theta & -2\sin(90) \\ 0\sin\theta & 2\cos(90 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

I think we use $-\theta$ IDK why but we do. But intuition telling me there's another way

for both



$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$

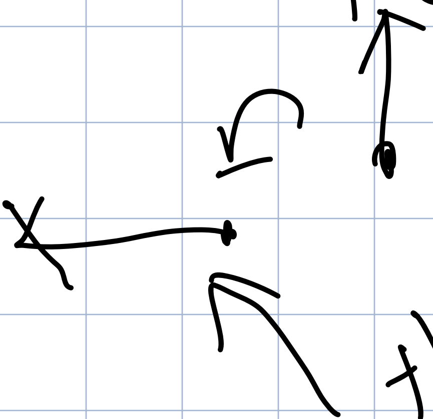
origin
and rotat

so

$\|v\| = 2$

$\theta = 0$

if we rotate
the vector 90



this is the
vector we
want in the
origin frame
for the other
neaw right?

so think $\sin\theta$ here
and $\cos\theta$ here

are what

we add.

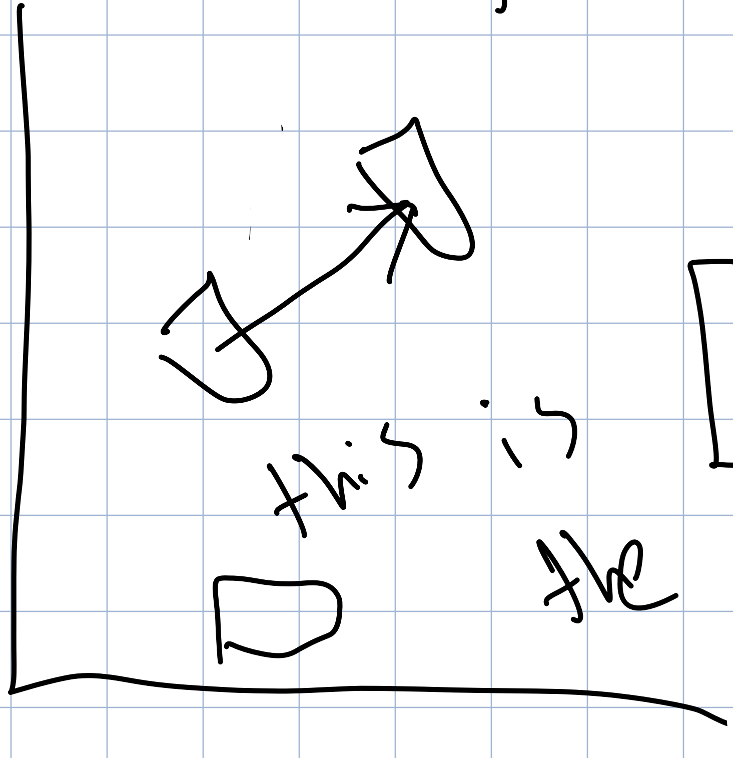
are I think that's

right b/c

sin(90), cos(90)

1 , 0 which

is the same



this is $\begin{bmatrix}2\\0\end{bmatrix}$ for

the vecto

but

$\begin{bmatrix}\sqrt{2}\\\sqrt{2}\end{bmatrix}$

for atom

we know

$$\begin{bmatrix}\sqrt{2}\\ \sqrt{2}\end{bmatrix}\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix} \|2\| = 2$$

and $\theta =$

$$= \begin{bmatrix}2\\ 0\end{bmatrix}$$

for the neeto 45 desrees

left

what if I jus say

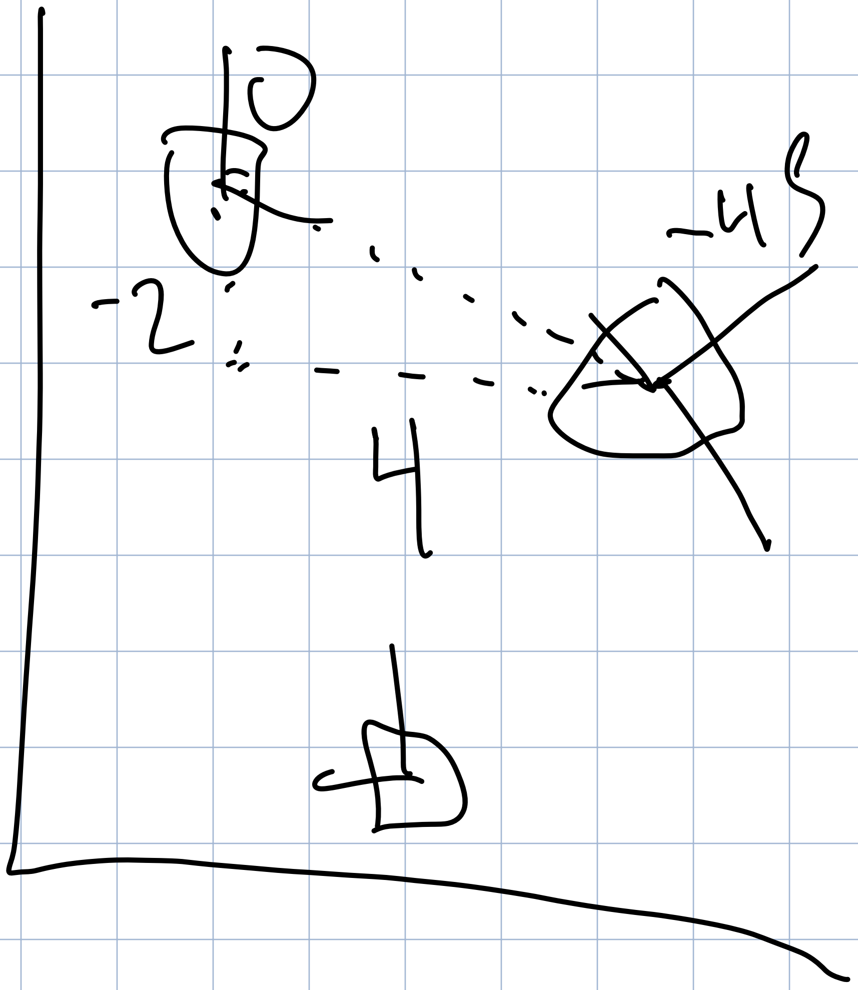
$\|2\|$ but $\theta = 90$

then once again

its in the atom

frame as $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

I don't know the heat 0 frame but this works.

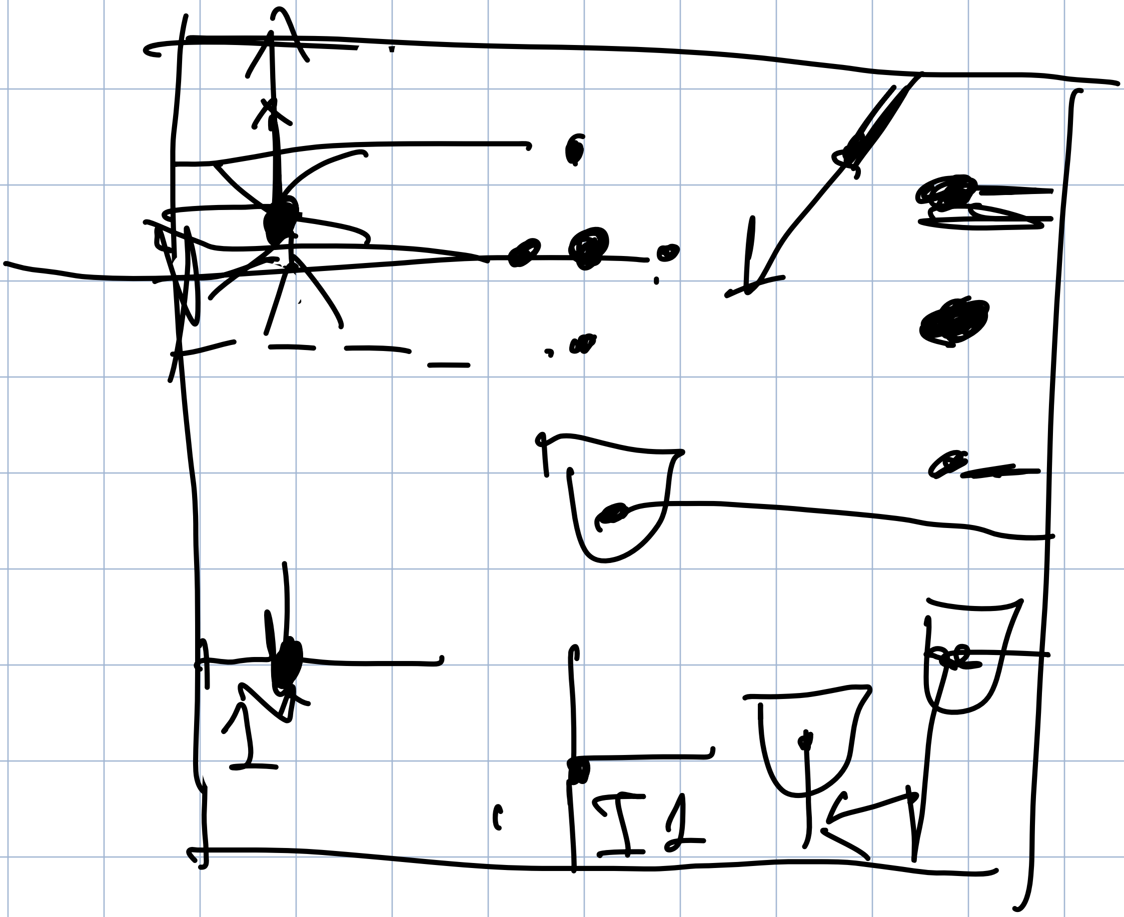


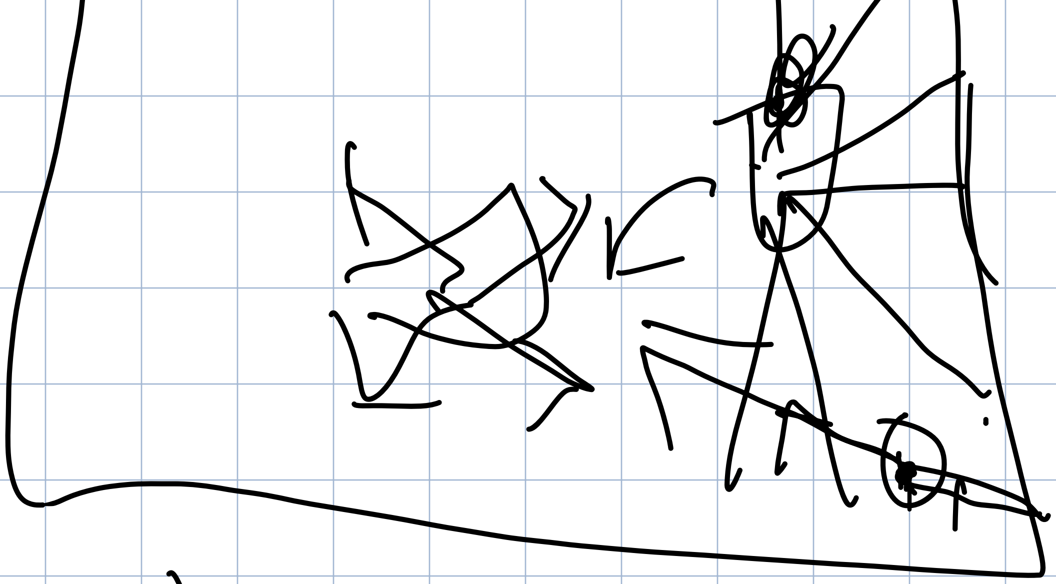
vector $\theta = \text{atan2}\left(\frac{\Delta y}{\Delta x}\right)$

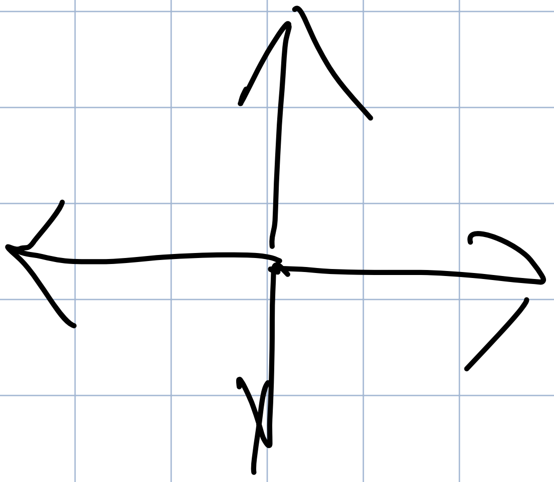
σ + diff(necto1, necto2)

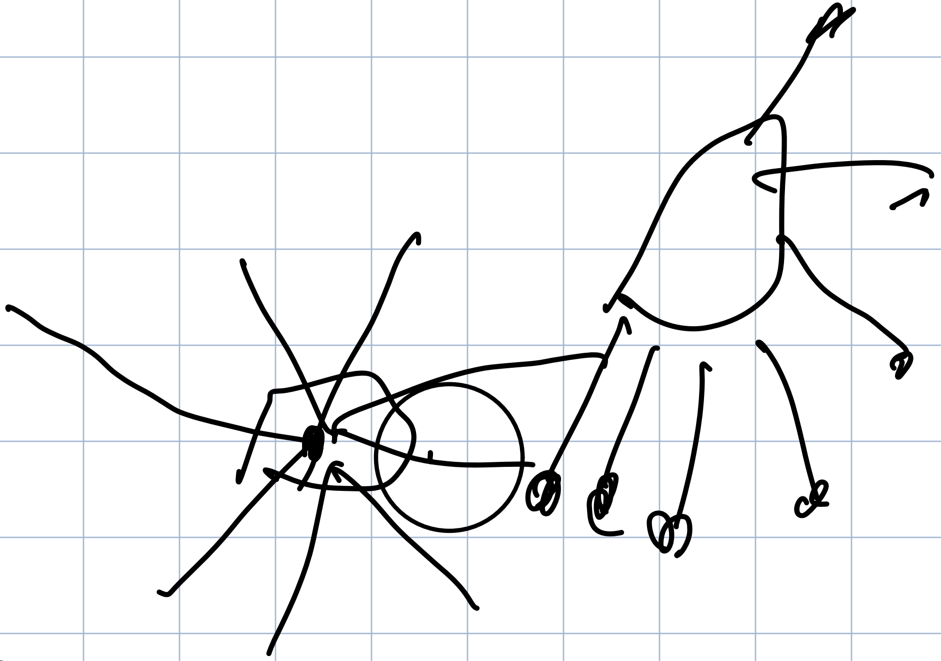
$= \sigma_{new}$

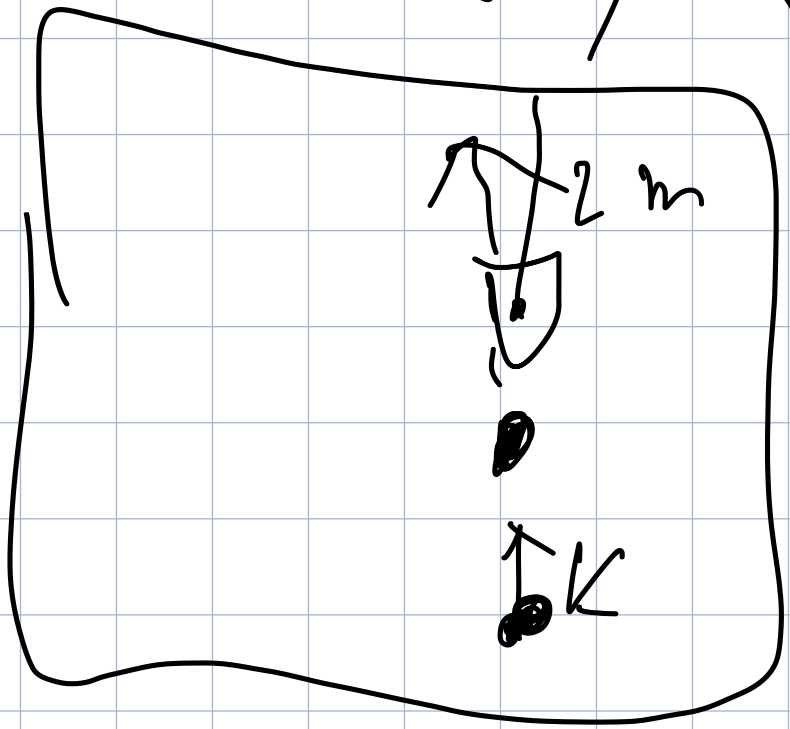
then sin cos







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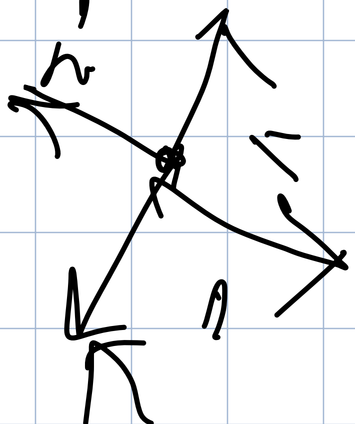
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